## 1. Battery in and E-stop bank B (LM5069-1, two FETs). Bank A, in series ahead of it, is block 7

Pack: 8 AWG pigtail pair to an XT90 (mates the pack XT90S), 1 BAT, 2 GND

0.5mR Bourns CSS4J-4026R-L500F 5W

## 2. Control rail and E-stop latch (no logic IC)

12.1 V, 11 mA. On the battery side, so regen through the body diodes keeps it up with the BMS open. Runs the latch
Fed through F1 with its own 1 uF, clamped by a 15 V TVS, reverse diode OUT to IN: block 10.

## 2a. Pi heartbeat watchdog: no heartbeat, no latch

## 3. Braking clamp: 56.6–56.8 V on, 55.8–55.9 V off (typ), 57.5 V worst case, into 2 x HS50 20R in parallel

## 4. Bus capacitors

235 uF, 80 V: 5 x Panasonic EEH-ZC1K470P hybrid, 36 mOhm, 10 x 10.2 mm, active line (rev B5: the APZ 68 uF is obsolete); the fifth is in block 10d. The drives have their own bulk. VBUS peaks about 58 V: 27 % margin. Regen overshoot about 0.8 V before the clamp acts.

## 5. 12 V converter and 12 V distribution: on the buck board (buck12), on the J24 pigtail (block 10)

## 5a. Brick soft start (ahead of the E-stop switch)

## 6. Limb ports: 2 mOhm shunt, own breaker (block 6a), meter, soldered pigtail pair. Pigtail fuses inline.

2mR 2512 CSS2H-2512K-2L00F 5W

L arm: 16 AWG pigtail pair, 1 +, 2 return

## 6a. Limb port breakers: TPS48111-Q1 + one FET per port, latch-off, enabled by RUN

## 7. E-stop bank A: second series bank, own driver, ahead of bank B

## 8. Spare 12 V outputs: on the buck board (split S1). SPARE_SW is enabled from RUN through block 5c

## 9. E-stop pressed: button NO contact to the Pi

## 10. Rev B2 protection parts (refs appended so earlier refs keep their numbers)

## 10b. Absorber close option (M3): 0R fitted, the rest DNP

## 10c. Rev B4 review parts

## 10d. Rev B5 parts

Rev B: a part goes in only if leaving it out would hurt. Parts list: hardware/power-board/REV_B_PART
**marengo**
Sheet: /
File: marengo_pdu_revb.kicad_sch
***Title: marengo power board rev B***
Size: A0 | Date: 2026-11-06 | **Rev: B-wip**
KiCad E.D.A. 10.0.8 | Id: 1/1